



PRODUCT REQUIREMENTS

Product: **Embergrove Tower Defense**
Category: Game Development (Browser / HTML5)
Document version 1.0

1. Product Overview

Embergrove Games Cafe is a fantasy themed tabletop board game cafe. It wants a short, playable browser game embedded on its website so guests have something fun to play on a phone or laptop while they wait for a table. The game is a single screen tower defense that runs with no installation, opening directly from one **index.html** file.

In a tower defense game, enemies travel along a fixed path from an entry point to an exit. The player stops them by placing defensive turrets on the buildable tiles beside the path. Each enemy that escapes costs one life; each enemy destroyed pays gold that funds more turrets and upgrades. The player wins by clearing all eight waves with at least one life left.

2. Goals and Success Criteria

PRODUCT GOALS

- A complete, self contained game in one folder.
- Loads instantly and runs fully offline.
- Clear, readable fantasy ember visual style.
- One satisfying core loop: defend, earn, upgrade.

DEFINITION OF DONE

- All eight waves are playable end to end.
- Win and lose states both reachable.
- Two turret types with at least one upgrade.
- Fixed palette and on screen HUD in place.

3. Key Parameters

Parameter	Value
Platform	Web browser, HTML / CSS / JavaScript, HTML5 Canvas
Entry point	index.html, no server and no build step
Canvas size	960 x 640 pixels
Grid	24 x 16 tiles at 40 x 40 pixels each
Starting lives	20
Starting gold	100
Number of waves	8
Turret types	2 minimum (Arbor Sentry, Ember Cannon)
Win condition	Clear wave 8 with at least 1 life
Lose condition	Lives reach 0

Scope note: No backend, database, accounts, payment, online multiplayer, or external paid service. Sound is optional. Everything ships inside one root folder and runs offline from local files.

4. Field and Path

The play field is a 24 by 16 tile grid. One continuous path runs from the left edge entry tile to the right edge exit tile and bends six times. Path tiles cannot be built on; every other tile is buildable grass.

Order	Waypoint (col, row)	Turn
1	(0, 8) entry	start at left edge
2	(5, 8)	turn up
3	(5, 3)	turn right
4	(12, 3)	turn down
5	(12, 12)	turn right
6	(18, 12)	turn up
7	(18, 6)	turn right
8	(23, 6) exit	leave at right edge

5. Enemies

Each enemy has health, a movement speed in pixels per second, and a gold bounty paid when destroyed. A small health bar above each enemy shrinks as it takes damage.

Enemy	Health	Speed (px/s)	Bounty	Role
Sprite	30	60	8	Basic enemy, the bulk of every wave
Wisp	20	120	12	Fast and fragile, pressures slow turrets
Brute	120	40	20	Slow and tough, needs heavy damage

6. Wave Schedule

Provide exactly eight waves. Enemies spawn one at a time, about 0.8 seconds apart, and the next wave begins automatically after a 3 second gap. Later waves add more enemies and tougher types than wave 1.

Wave	Sprites	Wisps	Brutes	Total	Theme
1	8	0	0	8	Warm up
2	12	0	0	12	More numbers
3	10	2	0	12	First fast enemies
4	14	0	3	17	First brutes
5	10	6	0	16	Speed pressure
6	12	0	5	17	Heavy pressure
7	14	8	4	26	Mixed assault
8	16	6	8	30	Final wave

7. Turrets

At least two turret types are required. A turret automatically aims at the nearest enemy inside its range and fires at its fire rate, removing its damage value from that enemy.

Turret	Cost	Range (px)	Fire rate (shots/s)	Damage	Best against
Arbor Sentry	50	100	3.0	5	Fast, low health enemies
Ember Cannon	120	140	0.8	30	Tough, slow brutes

8. Upgrades

Turret	Upgrade cost	Effect
Arbor Sentry	40	+3 damage and +20 px range
Ember Cannon	90	+15 damage and +25 px range

The player selects a placed turret and pays the upgrade cost to improve it. At least one upgrade level per turret is required.

9. Economy and Lives

Gold

- Start with 100 gold, shown on screen at all times.
- Destroying an enemy adds its bounty.
- Placing a turret subtracts its cost.
- Placement is blocked when gold is too low.

Lives

- Start with 20 lives, shown on screen at all times.
- An enemy reaching the exit removes one life.
- Reaching 0 lives ends the game in a loss.

10. Game States

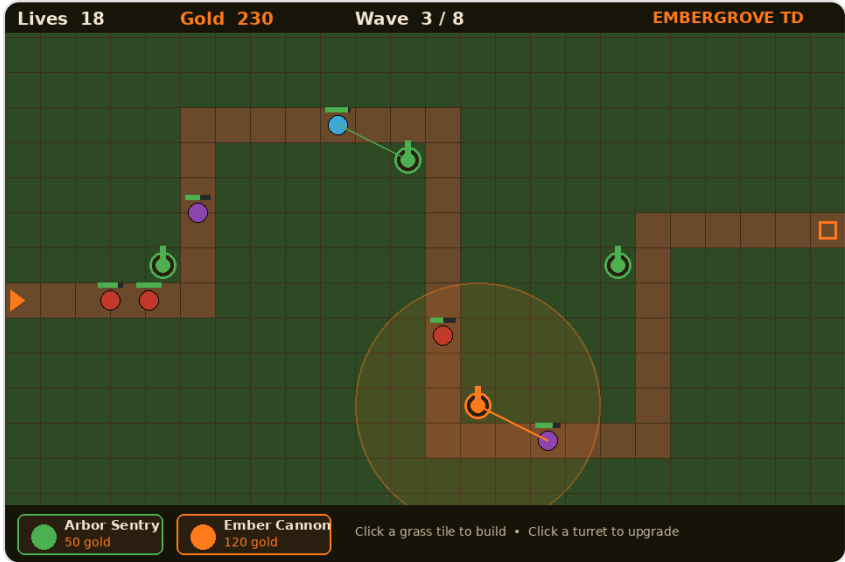
State	What happens
Start	Title screen with a clear Start button; the game begins on click.
Playing	Waves run, turrets fire, HUD shows lives, gold, and Wave X / 8.
Win	Shown when all eight waves clear with at least one life remaining.
Lose	Shown when lives reach zero; play stops.
Restart	Win and lose screens offer a restart that needs no page reload.

11. Visual Design

Use the fixed fantasy ember palette below across the field, HUD, turrets, and screens.

Background #1A1208	Path #6B4A2B	Grass #2E4A24	Accent / Fire #FF7A1A
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The image below shows the intended look of the screen during play: the winding path, turrets beside it, enemies with health bars, the top HUD, and the build bar.



12. Technical Requirements

Area	Requirement
Stack	HTML, CSS, JavaScript, HTML5 Canvas only
Run method	Open index.html in a modern browser
Loop	Animation loop via requestAnimationFrame
Offline	No network calls at runtime
Folder	index.html, styles.css, game.js, assets/, README.md

13. Deliverables and Acceptance

DELIVERABLES

- Root folder Embergrove_Tower_Defense.
- index.html entry point.
- styles.css and game logic in JavaScript.
- assets/ folder for any images.
- README.md with run steps and controls.

ACCEPTANCE SUMMARY

- Opens from index.html, runs offline.
- Correct canvas, grid, path, and palette.
- Eight waves, growing difficulty, HUD counters.
- Two turrets, placement, firing, one upgrade.
- Win, lose, and restart all work.

Out of scope: backend, database, accounts, payment, online multiplayer, paid services, and any required audio.